



UMT

University of Management and Technology, Lahore Campus

Semester Project: Spring 2026

Course Name:	Natural Language Processing	Course Code:	CC438	Credit Hours:	3
Course Instructor/s:	Mr. Waqar Ashiq	Program Name	BS Software Engineering		
Semester/ Section:	BSSE	Maximum Marks:			
Submission Deadline:	Monday 22 June, 2026	Moderator Signature			

Instructions:

- Understanding the problem is a part of the task.
- There is a ZERO tolerance policy for plagiarism. Tasks found with copy-pasted material or copied from fellows will be marked as ZERO.
- Submit your assignments on LMS within the given time frame. Emailed Tasks will not be accepted.

Submission Guidelines

- Submit **one ZIP file only**
- Work must be **individual (no group work allowed)**
- Upload through LMS only
- Ensure the project is **fully runnable**

CLO-4

Objective:

This task is designed to give students hands-on experience with modern NLP system design, focusing on **Retrieval-Augmented Generation (RAG)**, prompt engineering, and end-to-end deployment of intelligent chatbot systems using personal or custom datasets.

Task Description

You are required to design and implement a **RAG-based chatbot system** using your **own personal dataset**. The chatbot should be capable of answering user queries by retrieving relevant information from your dataset and generating contextual responses using an LLM.

Each student must build an **individual project**. Your system must include a **personalized chatbot identity (name-based chatbot)**, e.g., “AliBot”, “SaraAssistant”, etc.

Requirements

1. Dataset

- You must use **personal or custom-created data** (NOT generic datasets like Wikipedia dumps only).
- Data can include:
 - Personal documents including CV, Cover Letter
 - Semester Project information's
 - Research notes, projects, portfolio
 - PDFs, text files, or structured documents
- Data must be properly preprocessed and indexed for retrieval.

2. RAG System Implementation

Your system must include:

- Document loading and preprocessing
- Embedding generation
- Vector database (FAISS, ChromaDB, Pinecone, or similar)
- Retrieval mechanism
- LLM-based response generation
- Proper prompt engineering for context-aware answers
- History maintenance
- Build a working chatbot interface (CLI, Web app, or Streamlit preferred)
- Chatbot must have a **personal identity/name**
- It should respond using retrieved context + LLM generation (RAG pipeline)
- You can add agentic functionality in RAG

3. Deployment (Mandatory)

- The system must be **deployed on any platform**, such as:
 - Streamlit Cloud
 - Hugging Face Spaces
 - Render
 - Vercel (if applicable)
 - Local runnable deployment is NOT sufficient alone
- Deployment link must be functional at submission time

4. Code Submission

You must submit a **ZIP file** containing:

- Complete source code
- Requirements.txt (or environment setup file)
- README file (mandatory)
- Any dataset files used (or access instructions if large)
- Deployment instructions

Additional Note

Students may choose to implement **any other NLP/AI-related project idea** instead of the Personal RAG Chatbot, provided that:

- The project demonstrates a meaningful application of NLP, Generative AI, LLMs, Prompt Engineering, or related technologies.
- The project is implemented and submitted as an **individual effort**.
- The project is fully functional and executable.
- Complete source code, documentation, and deployment link (where applicable) are provided.

Students selecting an alternative project must ensure that the scope and complexity are comparable to the RAG-based chatbot project. Prior approval from the instructor is recommended for unique project ideas.